Milestone 2

Project goal

We aim to display how the profile of a dominant ATP player has changed from 1968 to the present. Rather than producing a single all-time ranking, our website reveals which features best characterize elite performance across the various eras of play. The intended audience ranges from casual tennis followers to coaches, equipment engineers, and sports regulators.

Live prototype

The website with the basic skeleton of the visualizations can be found at: https://com-480-data-visualization.github.io/nadal-prime/.

Core visualization

The first part is an introductory overview of the different eras of play, featuring historical context and the notable styles of play that dominated each period. We use bar charts to display the top-ranked players from each era by aggregated ranking points (see Figure 1).

The second part covers playstyle evolution, where we show relevant play statistics for each era through scatterplots (see Figure 2) and Sankey diagrams (see Figure 3).

Improvements on core visualization

The website will feature smooth scrolling between eras. We could also add a third section featuring visualizations across the entire timeline. One possibility is a parallel coordinates plot of different features (serve pts%, net pts%, rally length, aces%, etc.) for top players across eras, allowing trends to be read in a single view (see Figure 5). Complementing this, spider charts can overlay players from different eras on the same play statistics of for the top player against the average of the era (see Figure 4). Both plots would particularly benefit from consistent color coding across all the visualizations. Finally, trend lines for individual features would let readers track how each statistic has evolved over time.

All plots would be interactive include relevant information on hover.

Required tools

For the current prototype, the bar charts use d3-scale and d3-axis. The scatterplots add d3-zoom for panning and zooming. The Sankey diagrams are rendered as static SVG skeletons.

For the improved version, the Sankey diagrams will be replaced with data-driven layouts using the d3-sankey plugin. The parallel coordinates and spider charts can be implemented in D3, and smooth scrolling between sections will be handled by GSAP's ScrollTrigger.

Relevant lectures are "D3.js", “More interactive d3.js”, “Perception colors”, and “Storytelling”.

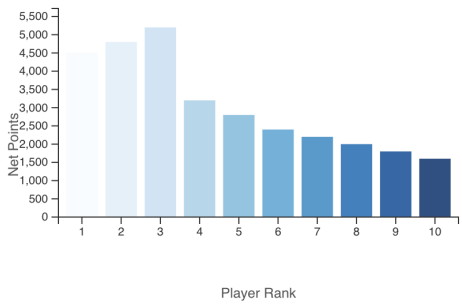

Figure 1: Bar chart showing the top 10 players by aggregated ranking points for a given era

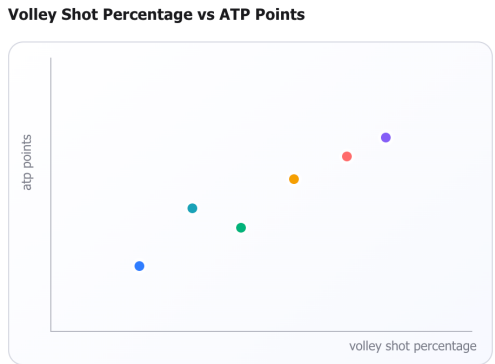

Figure 2: Scatter plot of volley shot percentage vs. ATP ranking points for players within a given era

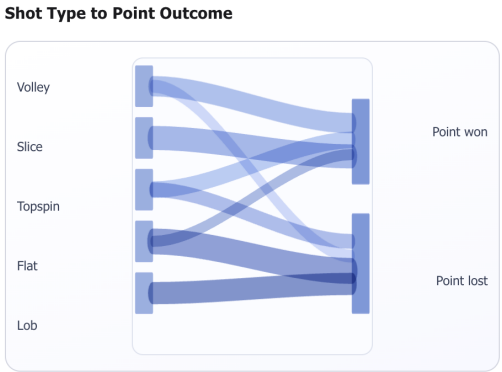

Figure 3: Sankey diagram skeleton showing shot type mapped to point outcome.

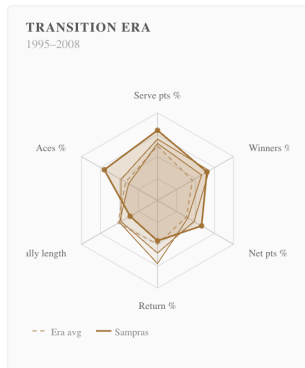

Figure 4: Spider chart example showing a top player's profile against the era average across six play statistics

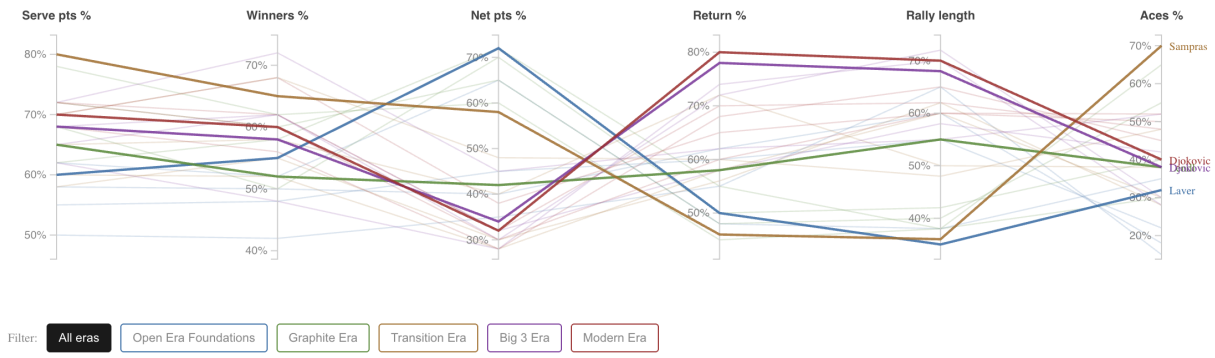

Figure 5: Parallel coordinates example showing player profiles across six statistics, color-coded by era, with era filtering